

Udiptaman Das

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PhD Candidate in Computer Science at the University of Missouri–Kansas City. My research centers on multimodal knowledge graphs (MMKGs) and trustworthy medical AI — integrating clinical text, imaging, and segmentation into structured, ontology-grounded knowledge for reasoning and decision support. Interests span multimodal reasoning, knowledge representation, and neuro-symbolic AI, with active collaborations across surgical oncology, airway modeling, and clinical decision-making.

TECHNICAL SKILLS

Languages & Tooling: Python, SQL, C++, Java, R, Bash; Docker, Git, PostgreSQL, MongoDB.

ML, LLMs & Agentic AI: PyTorch, TensorFlow, HuggingFace Transformers; LangChain, CrewAI, AutoGen; RAG, LoRA/QLoRA fine-tuning, uncertainty quantification.

Knowledge Graphs & Medical AI: Multimodal knowledge graphs (MMKG), RDFLib, Neo4j, SPARQL, OWL/RDFS, Protégé; ontology alignment (SNOMED CT, LOINC, RxNorm, UMLS, RadLex); medical imaging (segmentation, 3D/4D reconstruction); AWS (S3, Lambda, SageMaker), GCP.

EDUCATION

University of Missouri, Kansas City

PhD Candidate, Computer Science

Passed Comprehensive Examination — May 28, 2026

Kansas City, USA

Expected May 2027

University of Missouri, Kansas City

MS in Data Science and Analytics

Kansas City, USA

July 2024

Indian Institute of Technology

Foundational level degree in Programming and Data Science

Madras, India

Dec 2021

Assam Engineering College

BE in Electrical Engineering

Assam, India

Feb 2020

RESEARCH EXPERIENCE

University of Missouri, Kansas City

Doctoral Researcher (Advisor: Dr. Yugyung Lee) Aug 2023 – Present

Image-grounded vision knowledge graphs (VKG): Vision knowledge graphs that link medical imaging to structured clinical context for topology-aware analysis and explainable surgical-oncology decision support.

Abdominal oncology imaging-phenotype KG: Ontology-grounded knowledge graphs built from abdominal CT to enable explainable oncology reasoning over imaging phenotypes.

TrustKG — trustworthy clinical KG construction: Multi-agent LLM pipelines that convert unstructured clinical narratives into structured, ontology-aligned multimodal knowledge graphs, with retrieval grounding and uncertainty-aware validation.

Airway digital-twin MMKG (Project GENESIS — Children’s Mercy Hospital, ongoing): 4D (3D + temporal) segmentation-driven multimodal knowledge graphs of pediatric airway anatomy for digital-twin modeling and clinical decision support.

Large-scale corporate disclosure mining engine: Designed the ML stack for a financial-research collaboration with CUNY Brooklyn, extracting mission/strategy, talent-risk, and stock-compensation disclosures from SEC 10-K filings. Built a FIBO-guided, retrieval-first pipeline — hybrid retrieval (BM25 + dense embeddings with reciprocal rank fusion and cross-encoder reranking), NLI validation, and a LoRA-adapted quantized LLM for field extraction, with anomaly handling for auditability.

Community health web platform (Our Health Jackson County): Contributed to the development of [WeListen](#), a React-based web platform supporting regional health-disparity initiatives through community feedback and data collection.

WORK EXPERIENCE

Graduate Teaching Assistant, University of Missouri–Kansas City (Aug 2024 – May 2026) — C++ and Java course instruction, lab support, and homework grading.

Student Assistant, University of Missouri–Kansas City (Jan 2023 – Dec 2023) — Operational and inventory support across campus labs.

Data Science Intern, Deepneura Technologies (Mar 2022 – May 2022) — Object detection in gaming video and signature detection in document images.

Junior Data Engineer, Dibwas Websoft Technologies (May 2021 – Oct 2021) — Built and tuned ML models; SHAP-based KPI analysis and Python-based data visualization.

PUBLICATIONS

1. **Das, U.**, et al. “Ontology-Grounded Imaging Phenotype Knowledge Graphs for Explainable Oncology Reasoning from Abdominal CT.” Accepted at AMIA 2026 Annual Symposium.

2. Atmakuri, K. B., **Das, U.**, et al. “NS-CARE: Risk-Aware Neuro-Symbolic Clinical Decision Support for Safe Medication Recommendation from Electronic Health Records.” Accepted at AMIA 2026 Annual Symposium.

3. **Das, U.**, et al. “From Perception to Reasoning: Image-Grounded Knowledge Graphs for Topology-Aware Medical Analysis in Abdominal CT.” 1st Workshop on Medical Reasoning (MedReasoner), CVPR 2026.

4. **Das, U.**, et al. “Clinical Knowledge Graph Construction and Evaluation with Multi-LLMs via Retrieval-Augmented Generation.” arXiv:2601.01844.

5. Ho, D. H., **Das, U.**, et al. “Leveraging Multi-Agent Systems and Large Language Models for Diabetes Knowledge Graphs.” IEEE Big Data Conference — 4th Workshop on Knowledge Graphs and Big Data.

Under Review

6. **Das, U.**, et al. “TRUST-KG: Trustworthy Retrieval-Grounded Clinical Knowledge Graph Construction from Unstructured Oncology Narratives.” Under review, IEEE Big Data 2026.

7. **Das, U.**, et al. “Ontology-Aligned Vision Knowledge Graphs for Surgical Oncology Decision Support.” Under review, AAAI 2027.

8. **Das, U.**, et al. “Multimodal Knowledge Graphs for 4D Airway Segmentation and Digital-Twin Modeling.” Under review, WACV.